UNIVERSITY OF KARACHI

Department of Computer Science (UBIT)

LAB NO. 04

Operating Systems (SE-458)

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Date of Submission: 20 October 2026

LAB NO. 04: LINUX FILE PERMISSIONS AND OWNERSHIP

Objective

To gain practical experience with Linux file system permissions, user and group ownership management, permission modification using chmod in numeric and symbolic modes, and configuring default file creation masks with umask.

PART A: FILE PERMISSIONS AND COMMAND REFERENCE

1. Users, Owners and Groups

Linux supports multiple user accounts. Permissions control which users can read, modify or execute files and access directories. Every file and directory has an owner and an associated group.

- **Owner (u):** The user account that owns the item. This is usually the account that created it, although ownership can be changed.
- **Group (g):** The group associated with the item. Its permissions apply to users who belong to that group.
- **Others (o):** Users who are neither the owner nor members of the associated group.
- **All (a):** A shorthand used by chmod to select owner, group and others together. It is not a fourth permission category.

Useful identity commands:

- **whoami** displays the effective username.
- **id** displays user and group IDs and group memberships.
- **id -gn** displays the current user's primary group name.
- **id -Gn** displays the names of the groups the user belongs to.

2. Read, Write and Execute

Permissions on Files

- **Read (r):** Allows the contents of a file to be viewed.
- **Write (w):** Allows the contents to be modified.
- **Execute (x):** Allows the file to be executed directly, provided it contains a valid program or script.

Permissions on Directories

- **Read (r):** Allows the names of entries inside the directory to be listed.

- **Write (w):** Allows entries to be created, deleted or renamed, normally together with execute permission.
- **Execute (x):** Allows the directory to be entered and its entries to be accessed by name.

3. Inspecting Permissions with ls

The ls -l command displays permissions and ownership information.
Example permission string: -rwxr-xr--

- The first character shows the type: - for a regular file, d for a directory, or l for a symbolic link.
- The next three characters, rwx, belong to the owner.
- The following three characters, r-x, belong to the group.
- The final three characters, r--, belong to others.

Useful commands:

- **ls -l report.txt** displays detailed information about the file.
- **ls -ld data** displays the data directory's own permissions.
- **ls -la** includes hidden entries.
- **ls -n** displays numeric user and group IDs.

4. Numeric Permissions

Numeric permissions use three digits in the order owner, group and others. Each digit is calculated by adding the required permission values:

- **Read = 4.**
- **Write = 2.**
- **Execute = 1.**

Common permission modes:

- **644:** Owner can read and write. Group and others can read.
- **755:** Owner has all permissions. Group and others can read and execute.
- **777:** All three categories have read, write and execute.

5. chmod: Change Mode

chmod changes the permissions of an existing file or directory.
Numeric Mode Examples:

- **chmod 644 report.txt** gives the owner read and write access, and gives everyone else read access.
- **chmod 755 script.sh** gives the owner full permissions and gives group members and others read and execute access.

Symbolic Mode:

- **chmod u+x script.sh** adds execute permission for the owner.

- **chmod g+w report.txt** adds write permission for the group.

6. chown: Change Owner

chown changes a file's owner. Syntax: chown [options] owner[:group] file

7. chgrp: Change Group

chgrp changes the group associated with a file or directory.

8. umask: File Creation Mask

umask specifies which permission bits are removed when new files and directories are created.

PART B: PRACTICAL WORK AND CLASS EXERCISES

Lab Task 1: Create a Permission Environment

```shell
mkdir linux_permission_lab
cd linux_permission_lab
touch report.txt
touch script.sh
mkdir project
ls -l
ls -ld project
```

Output:



```
mahad@ubunto-dev ~/OS-Lab/linux_permission_lab
❯ ls -l
total 4
drwxrwxr-x 2 mahad mahad 4096 Oct 21 17:04 project
-rw-rw-r-- 1 mahad mahad    0 Oct 21 17:04 report.txt
-rw-rw-r-- 1 mahad mahad    0 Oct 21 17:04 script.txt

mahad@ubunto-dev ~/OS-Lab/linux_permission_lab
❯ ls -ld project
drwxrwxr-x 2 mahad mahad 4096 Oct 21 17:04 project
```

Lab Task 2: Change Permissions Using Numeric Mode

Shell

```
ls -l
chmod 600 report.txt
ls -l report.txt
chmod 644 report.txt
ls -l report.txt
chmod 755 script.sh
ls -l script.sh
chmod 700 project
ls -ld project
```

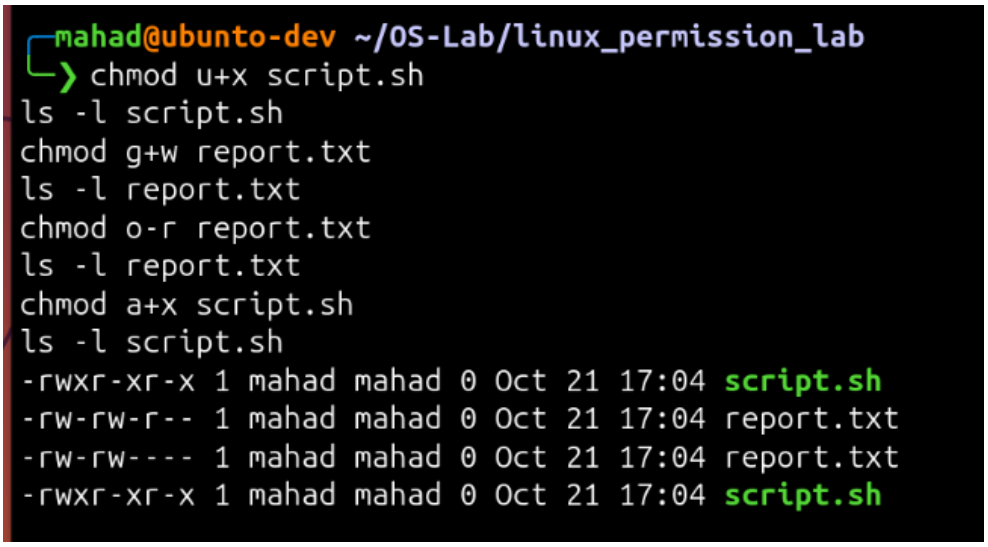
Output:

```
mahad@ubunto-dev ~/OS-Lab/linux_permission_lab
❯ ls -l
chmod 600 report.txt
ls -l report.txt
chmod 644 report.txt
ls -l report.txt
chmod 755 script.sh
ls -l script.sh
chmod 700 project
ls -ld project
total 4
drwxrwxr-x 2 mahad mahad 4096 Oct 21 17:04 project
-rw-rw-r-- 1 mahad mahad    0 Oct 21 17:04 report.txt
-rw-rw-r-- 1 mahad mahad    0 Oct 21 17:04 script.txt
-rw------- 1 mahad mahad 0 Oct 21 17:04 report.txt
-rw-r--r-- 1 mahad mahad 0 Oct 21 17:04 report.txt
chmod: cannot access 'script.sh': No such file or directory
ls: cannot access 'script.sh': No such file or directory
drwx------ 2 mahad mahad 4096 Oct 21 17:04 project
```

Lab Task 3: Change Permissions Using Symbolic Mode

```shell
chmod u+x script.sh
ls -l script.sh
chmod g+w report.txt
ls -l report.txt
chmod o-r report.txt
ls -l report.txt
chmod a+x script.sh
ls -l script.sh
```

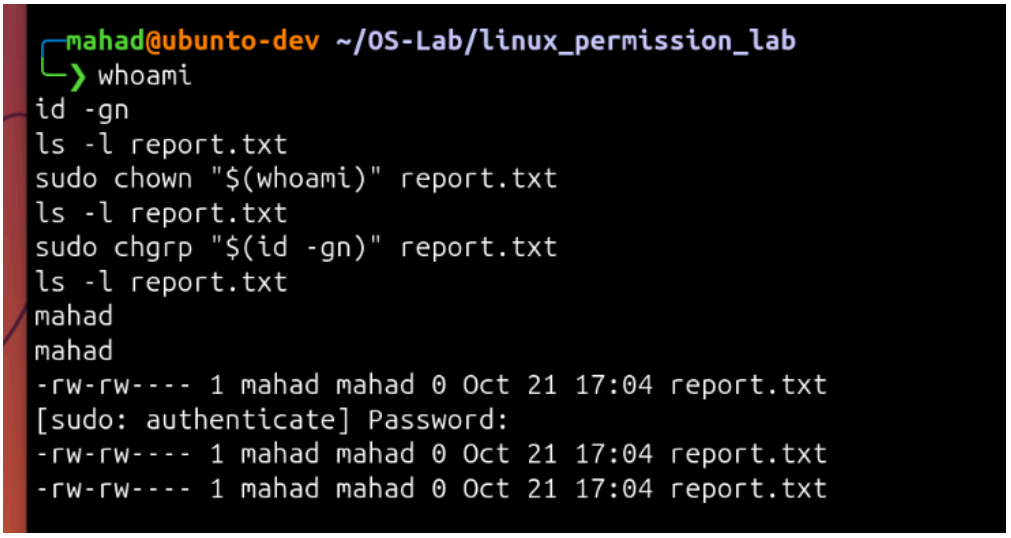
Output:



Lab Task 4: Manage File Ownership Using chown and chgrp

```shell
whoami
id -gn
ls -l report.txt
sudo chown "$(whoami)" report.txt
ls -l report.txt
sudo chgrp "$(id -gn)" report.txt
ls -l report.txt
```

Output:



Lab Task 5: Investigate Default Permissions Using umask

```shell
lab4_original_umask=$(umask)
umask
touch file1.txt
mkdir dir1
ls -ld file1.txt dir1
umask 027
umask
touch file2.txt
mkdir dir2
ls -ld file2.txt dir2
umask "$lab4_original_umask"
```

Output:

```
mahad@ubunto-dev ~/OS-Lab/linux_permission_lab
❯ lab4_original_umask=$(umask)
umask
touch file1.txt
mkdir dir1
ls -ld file1.txt dir1
umask 027
umask
touch file2.txt
mkdir dir2
ls -ld file2.txt dir2
umask "$lab4_original_umask"
0002
drwxrwxr-x 2 mahad mahad 4096 Oct 21 17:19 dir1
-rw-rw-r-- 1 mahad mahad    0 Oct 21 17:19 file1.txt
0027
drwxr-x--- 2 mahad mahad 4096 Oct 21 17:19 dir2
-rw-r----- 1 mahad mahad    0 Oct 21 17:19 file2.txt
```

Lab Task 6: Display Basic System Information

```shell
who
whoami
tty
date
cal
```

Output:

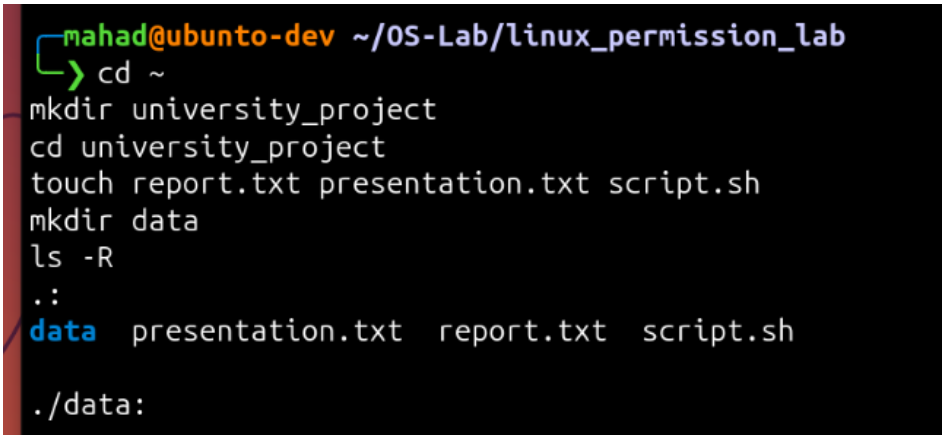
```
mahad@ubunto-dev ~/OS-Lab/linux_permission_lab
❯ who
whoami
tty
date
cal
1000     seat0        2026-10-21 10:09
1000     tty2         2026-10-21 10:09
mahad
Wed Oct 21 05:19:37 PM PKT 2026
    October 2026
Su Mo Tu We Th Fr Sa
              1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31
```

CLASSWORK: UNIVERSITY PROJECT TEAM

Task 1: Create the Directory Structure

```shell
cd ~
mkdir university_project
cd university_project
touch report.txt presentation.txt script.sh
mkdir data
ls -R
```

Output:



Task 2: Display Permissions

```shell
ls -l
ls -ld data
```

Output:

```
mahad@ubunto-dev ~/university_project
❯ ls -l
ls -ld data
total 4
drwxrwxr-x 2 mahad mahad 4096 Oct 21 17:20 data
-rw-rw-r-- 1 mahad mahad    0 Oct 21 17:20 presentation.txt
-rw-rw-r-- 1 mahad mahad    0 Oct 21 17:20 report.txt
-rw-rw-r-- 1 mahad mahad    0 Oct 21 17:20 script.sh
drwxrwxr-x 2 mahad mahad 4096 Oct 21 17:20 data
```

Task 3: Change report.txt Permissions to 644

```shell
chmod 644 report.txt
ls -l report.txt
```

Output:

```
mahad@ubunto-dev ~/university_project
❯ chmod 644 report.txt
ls -l report.txt
-rw-r--r-- 1 mahad mahad 0 Oct 21 17:20 report.txt
```

Task 4: Change script.sh Permissions to 755

```shell
chmod 755 script.sh
ls -l script.sh
```

Output:

```
mahad@ubunto-dev ~/university_project
❯ chmod 755 script.sh
ls -l script.sh
-rwxr-xr-x 1 mahad mahad 0 Oct 21 17:20 script.sh
```

Task 5: Change data Directory Permissions to 700

```shell
chmod 700 data
ls -ld data
```

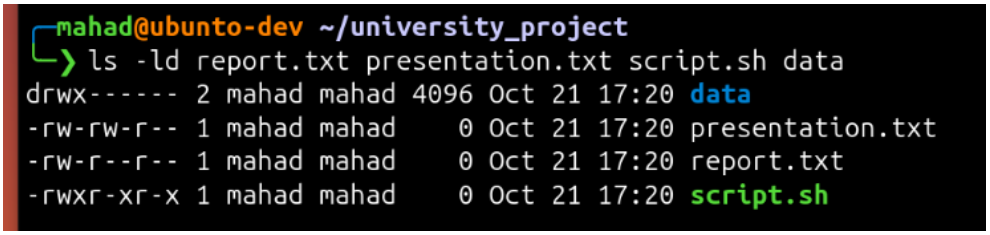
Output:



Task 6: Inspect the Owner and Group of Each Item

```shell
ls -ld report.txt presentation.txt script.sh data
```

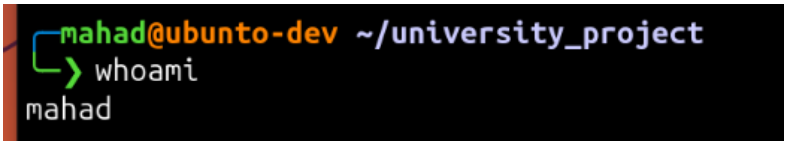
Output:



Task 7: Display the Current User

```shell
whoami
```

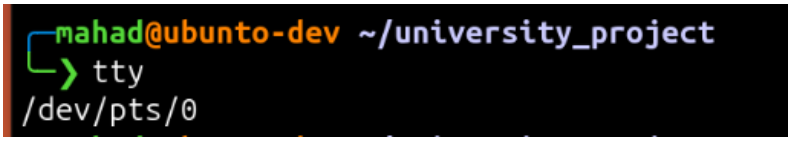
Output:



Task 8: Display the Current Terminal

```
Shell
tty
```

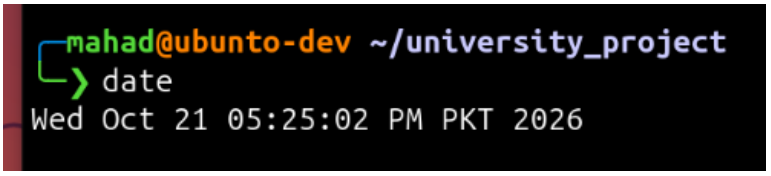
Output:



Task 9: Display the Current Date

```
Shell
date
```

Output:



Task 10: Display the Calendar

```
Shell
cal
```

Output:

